

EUNSONG KOH

Vancouver, BC | esongkoh@gmail.com | linkedin.com/in/songkoh | songk.xyz | github.com/eunsongkoh

EDUCATION

Simon Fraser University

Bachelor of Science, Computer Science

Expected Graduation April 2027

Vancouver, BC

- **Relevant Coursework:** Database Systems, Systems Programming, Networking, Computational Data Science, Data Structures & Algorithms
- **Extracurriculars:** SFU DNS Club Dir. Communications & Multimedia, SFU WiCS Mentor & Workshop Lead

EXPERIENCE

Machine Learning Engineer Intern

Amazon

May 2026 - Jul 2026

Vancouver, BC

- Incoming Summer 2026

Applied Research Engineer

Safety CLI Cybersecurity | Pre-Series A (Backed by First Round Capital & Kearny Jackson)

Jan 2026 - Apr 2026

Vancouver, BC

- Improved LLM-based vulnerability classification precision from **28% to 78%** through prompt optimization and benchmarking across multiple decoder models; experimented with **BERT-based** fine-tuning and hyperparameter optimization for classification evaluation
- Reduced LLM inference costs by **40%** through model selection and chunking strategy optimizations, while building automated evaluation and experiment tracking pipelines using **LangSmith/LangChain**
- Redesigned sequential ECS vulnerability analysis into a **distributed graph-processing pipeline** on AWS, improving throughput by **5x** and reducing compute costs by **50%**
- Developed an internal vulnerability triage platform with automated classification and analyst review workflows with **React**, reducing projected manual review workload by **90%+**

Undergraduate Researcher

Reliable Systems Lab | Simon Fraser University

May 2025 - Apr 2026

Vancouver, BC

- Co-authored research on **LLM-based automated vulnerability discovery and repair** supervised by Dr. Steven Ko
- Applied advanced **prompt engineering** techniques to improve model-guided detection by 10%
- Developed **benchmarking pipelines** for Rust vulnerabilities using LangChain and LangSmith, increasing detection coverage and reliability

Software Engineer Intern

Safety CLI Cybersecurity | Pre-Series A (Backed by First Round Capital & Kearny Jackson)

May 2025 - Dec 2025

Vancouver, BC

- Built a **full-stack vulnerability analysis platform** using **FastAPI + React**, leading 90% of frontend development; accelerated security reviews 4x and improved team productivity 16%
- Engineered and productionized a **high-performance call-graph traversal algorithm** for vulnerability detection (50k+ nodes in under 1 second) in the data engine, enabling scalable, real-time system analysis
- Prototyped and evaluated decoder and **Hugging Face Transformer** models on unstructured data, applying **distributed training on AWS EC2** with **Accelerate** and **Ray Train/Tune** to reduce training time by 25%; Developed a **ML-powered data licensing and evaluation pipeline** using **LangChain** and **LangSmith**, achieving 92% evaluation accuracy

Software Developer & Product Manager

Hope Health Action | Simon Fraser University

Jan 2025 - Apr 2025

Vancouver, BC

- Designed and implemented a **user history tracking endpoint** and corresponding **React/React Native dashboard** as part of a research initiative under Dr. Brian Fraser, developing a NGO-backed app on the **Google Play Store**
- Redesigned core **Django** data models with **PostgreSQL** and refactored backend and **React/React Native** systems, improving query efficiency by **15%** and frontend performance and maintainability by **13%**
- Authored **PRDs**, designed **Figma** prototypes, and collaborated with developers and clients to deliver **3 new features** on schedule

Software Engineer Intern

Kelowna Software Ltd.

May 2024 - Aug 2024

Remote

- Refactored and optimized legacy code, increasing **performance by 5%**, improving **readability by 15%**, and enhancing maintainability and test coverage (+20%) using **XUnit** and **Jasmine** across internal and client software
- Automated **large-scale ML data pre-processing** using **Python**, **Pandas**, and **Dask**, reducing data sorting and transformation time by **70%**
- Developed back-end systems in **C#/.NET Core** and client-facing features with **TypeScript/Angular** for **1000+** users

PROJECTS

AutoJob | AWS Step Functions, DynamoDB, Bedrock, EventBridge, GitHub Actions

- Built a **serverless job tracking platform** with AWS Step Functions and DynamoDB, creating scalable pipelines for automated notifications and state management
- Integrated **AWS Bedrock** for relevance-based classification, **EventBridge** for scheduled notifications, and automated **CI/CD pipelines with GitHub Actions** for reliable large-scale execution

TECHNICAL SKILLS

Languages: C, C++, C#, Python, Java, SQL, JavaScript, TypeScript, Swift, HTML/CSS

Frameworks/Libraries: .NET, Angular, React.js, Next.js, Redux, Django, JUnit/XUnit, Pandas, Bootstrap, TailwindCSS, SwiftUI

Developer Tools/Technologies: AWS, Azure, LangChain, HuggingFace, MongoDB, Docker, Git, Github Actions, Figma